Pairs with certain difference

Difficulty: **Easy** Accuracy: **63.41%** Submissions: **43K+** Points: **2**

Given an array of integers and a number k, the task is the find maximum pair sum with the following conditions on the pairs.

- Pair difference should be less than k.
- Pairs should be disjoint. For example if (x, y) is a result pair, then neither x nor y should appear in any other result pair.
- Sum of p pairs means sum of 2p elements in the result.

If no valid pairs can be formed, return 0.

Examples:

Input: arr[] = [3, 5, 10, 15, 17, 12, 9], K = 4
Output: 62
Explanation :
The valid disjoint pairs with difference less than K are:
(3, 5), (10, 12), (15, 17)
The maximum sum obtained from these pairs is:
3 + 5 + 10 + 12 + 15 + 17 = 62
An alternative pairing could be:
(3, 5), (9, 12), (15, 17)


```python
class Solution:

    def sumDiffPairs(self, arr, k):
        arr.sort()
        ans = 0
        i = len(arr) - 1

        while i > 0:
            if arr[i] - arr[i - 1] < k:
                ans += arr[i] + arr[i - 1]
                i -= 2
            else:
                i -= 1
        return ans
```